

# novelty-theory-lean:

## Fixed Generation Without Fixed Explanatory Closure

Machine-Checked Generative Systems, Stratified Provability, and Crown Packaging

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<https://github.com/novaspivack/novelty-theory-lean>

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### Abstract

**First sentence (existential flagship).** There exists a concrete deterministic generative system on  $\mathbb{N}$  (the library’s canonical counter) and a packaged sentence universe such that a family of *structurally certified* sentences exhibits strict proof-theoretic ascent under an explicit depth calculus, while generator truth on the same counter coexists with this phenomenon in a single bundled statement.

**Second sentence (class-transfer strengthening).** Beyond a single model artifact, the library constructs an admissible class of aligned  $\mathbb{N} \rightarrow (\text{Bool} \times \mathbb{N})$  generators for which the same enum-crown gap template transfers while the Bool observation disagrees everywhere—so the phenomenon is not tied to one carrier tag by construction.

**Third sentence (non-finality and terminality defeat).** The generalized summit also packages scoped non-finality of an explanatory regime diagram (as formalized) and a formal future-defeat property for bounded terminality-style claims on the canonical counter, alongside organization obstructions stated at bounded multiscale height.

**Trust boundary.** No literal universal over all raw admissible-interface rows, and no universal infinite paradigm shift for every singleton-within regime family, is claimed or available: such universal glosses are *false* under the formalized predicates and are explicitly refuted by obstruction lemmas in the artifact. All positive theorems are scoped to the definitions exported from the LEAN 4 library [4]. The formalization is zero-~~sorry~~ on the packaged library path and builds against Mathlib [3], toolchain in the repository pin [2].

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## 1 Premises, trust boundary, and what can still be attacked

### Where disagreement can (and cannot) live

**Formal spine.** All mathematical claims in this paper, when stated as theorems, refer to definitions and proofs in the publicly buildable LEAN 4 library [4]. Section A lists a theorem-to-prose map; Section B shows the informal layer graph.

**Premise coordinates (for external readers).**

- **Proof theory vs. metaphysics.** “Strict  $\vdash$  /  $\not\vdash$ ” is internal to the Model C depth calculus formalized in the library, not a claim about all possible logics or all physical universes.
- **Generation  $\neq$  explanation  $\neq$  provability.** Section 4 keeps these layers explicit; collapsing them in prose would overstate the formal result.
- **Scoped admissibility.** Class-transfer results quantify over the *constructed* aligned carriers used in the proofs—not an unqualified “all mathematical dynamical systems” statement.
- **Blocked literal universals.** Two universal templates are *refuted* in-library (see Section 3): raw interface diagonals without further scoping, and infinite paradigm chains for every singleton-within regime family.

**Keywords:** interactive theorem proving, LEAN 4, formalization, generative systems, stratified provability, explanatory regimes, anti-collapse, crown packaging

## 2 Introduction (outline)

**Problem coordinate.** *[Draft: fixed generation without fixed explanatory closure; observation/simulation need not explanatory reduction.]*

**Contribution (frozen flagship).** The primary named package is `NoveltyTheory.Summits.GeneralizedFin` supporting roles for earlier summits and `FinalCrownPackage` without coequal headline status.

**Roadmap.** Section 3: boundaries first. Section 4: generation / explanation / provability. Section 5: why the crown is not a mention-bound artifact. Section 6: genotype/phenotype, computational irreducibility, emergence rhetoric. Section 7: informal tour of the calculus. Section 10: limitations and future frontier (*richer phase-at-sentence modalities, broader carriers*).

### 3 Blocked universal templates (write boundaries before the intro polish)

**Raw admissible interface.** *[Draft: cite obstruction lemmas: trivial interfaces can make diagonal rows exhaustive; honest packaging uses signature-scoped or bundled admissibility.]*

**Singleton-within regime families.** *[Draft: empty regime / obstruction to universal infinite paradigm chain.]*

### 4 Three levels of novelty (do not let readers blur these)

**Generation (dynamics / trace level).** *[Draft.]*

**Explanation (regime / phase / organizational diagrams).** *[Draft.]*

**Provability and expressibility (sentence height / Model C).** *[Draft.]*

### 5 Anti-collapse: what the crown is demonstrably not

**Not only geOutput.** *[Draft: structural witness, enum crown, structural v2 / hist bundles.]*

**Not only trace-ridge atoms.** *[Draft.]*

**Not only mentionBound stratification.** *[Draft: paired sentences at same stratum, distinct formulas.]*

**Not a single-model artifact (Bool-tagged class).** *[Draft: link NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized\_final\_crown\_not\_model\_artifact.]*

### 6 Comparison with nearby narratives

**Genotype / phenotype and simple adequacy readings.** *[Draft keyed to observation  $\nrightarrow$  explanatory reduction lemmas.]*

**Computational irreducibility [5] and Rule 110 [1].** *[Draft: conceptual kinship vs. formal claim mismatch—this paper is proof-theoretic and regime-organizational, not a cellular automaton undecidability export.]*

**“Emergence” rhetoric.** *[Draft: only what LEAN 4 exports.]*

### 7 Formal setting (informal)

**Generative systems and the canonical counter.** *[Draft: states, step, trace, output.]*

**Sentences, semantics, and ProvesAt.** *[Draft: Model C bridge; depth as a resource.]*

**Regimes, phases, organization.** *[Draft: explains, bounded organization predicates.]*

## 8 Flagship generalized package (prose target)

[Draft: restate the six `generalized_final_crown_*` theorems informally; point to Appendix A for exact names and boundaries.]

## 9 What this paper does *not* prove (one-page discipline)

- Not: every deterministic system exhibits the strict-gap pattern.
- Not: every explanatory formalization fails globally.
- Not: every organization must be infinite (only bounded packaged obstructions as stated).
- Not: metaphysical anti-determinism or backward causation.
- Not: a universal diagonal over all raw interface rows, or a universal infinite paradigm tower.

## 10 Discussion and future frontier

[Draft: richer phase-at-sentence without `Set ℕ` encoding; extra modalities; additional carriers—explicitly “new program” beyond the frozen tranche.]

## 11 Conclusion

[Draft: freeze claim surface; point to repository build instructions.]

**Reproducibility.** From the repository root, `lake build` compiles `NoveltyTheory`; the paper bundle lives under `paper/`.

## A Theorem-to-prose map

| Informal claim<br>(draft)                                                                                                                                                                                                                             | Lean theorem                                                                                                       | Assumptions /<br>scope                                                                                                                        | Interpretation<br>boundary                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| There is a uniform family of sentences that are certified in a <i>structural</i> sense (generator-structural or retro-structural v2) and exhibit a strict proof-theoretic gap ( $\vdash$ at depth $n+1$ but not at $n$ ) under the packaged calculus. | <code>NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_iterated_structural_ascent</code> | Model C depth calculus ( <code>ProvesAt</code> ); canonical counter <code>natCounter</code> ; predicates <code>IsStructuralSentence v2</code> | Not a claim about <i>every</i> deterministic system or every logic—only the formalized generative system and sentence universe in the library. |

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Table 1 – continued

| Informal claim                                                                                                                                                                                                   | Lean theorem                                                                                                         | Assumptions                                                                                                                                                                      | Boundary                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| No fixed finite explanatory stratum (as formalized via regimes) exhausts the phenomenon; bounded multiscale organization cannot globally totalize a standard future diagram of strict N pairs.                   | <code>NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_no_fixed_structural_stratum</code>  | <code>Phase, regimeUpto</code> , explains packaging; <code>AdmissibleBoundedOrganization</code> class <code>TotalizesStandardFuture</code> predicates; not a obstruction lemmas. | Scoped to the formal <code>OrganizationV2 / Organization</code> class; not a metaphysical claim that “all organizations” fail.           |
| A formal <i>future defeat</i> property holds for the canonical counter (forbidding a terminality-style universal bound on observable outputs).                                                                   | <code>NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_future_defeat_of_terminality</code> | <code>FutureDefeat</code> on <code>natCounter</code> as defined in <code>StratifiedFinality</code> .                                                                             | This is a theorem about the <i>defined</i> terminality predicate and the canonical generator, not empirical open-endedness in the world. |
| Generator-truth linkage on the canonical counter coexists with a structural strict-ascent family (trace-equality atoms $\wedge$ structural ascent family in one bundle).                                         | <code>NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_generator_truth_family</code>       | <code>IsGeneratorStructuralHoldsAt natCounter</code> , <code>ProvesAt</code> gap for the structural-v2 family.                                                                   | <code>Bundles</code> two phenomena; does not assert that <i>every</i> true-on-counter sentence is structurally certified.                |
| The enum-crown gap pattern is compatible with an admissible <i>class</i> of aligned carriers (two concrete aligned systems on $\mathbb{N} \rightarrow (\text{Bool} \times \mathbb{N})$ with opposing Bool tags). | <code>NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_class_transfer</code>               | <code>ProvesAtCarrier</code> , <code>AlignsCarrier</code> , constructed <code>trueBoolProdNat / altBoolProdNat</code> pair; enum witness pullback.                               | <b>Existential / constructed class:</b> not “for all carriers” or “for all tags.”                                                        |
| The library exhibits two trace-distinct aligned systems (not a single-model artifact for the Bool-tagged transfer story).                                                                                        | <code>NoveltyTheory.Summits.GeneralizedFinalCrownPackage.generalized_final_crown_not_model_artifact</code>           | Same transfer construction as previous row; inequality of traces at some stage.                                                                                                  | Strengthens honesty of the class-transfer headline; still not a universal across all mathematical generators.                            |

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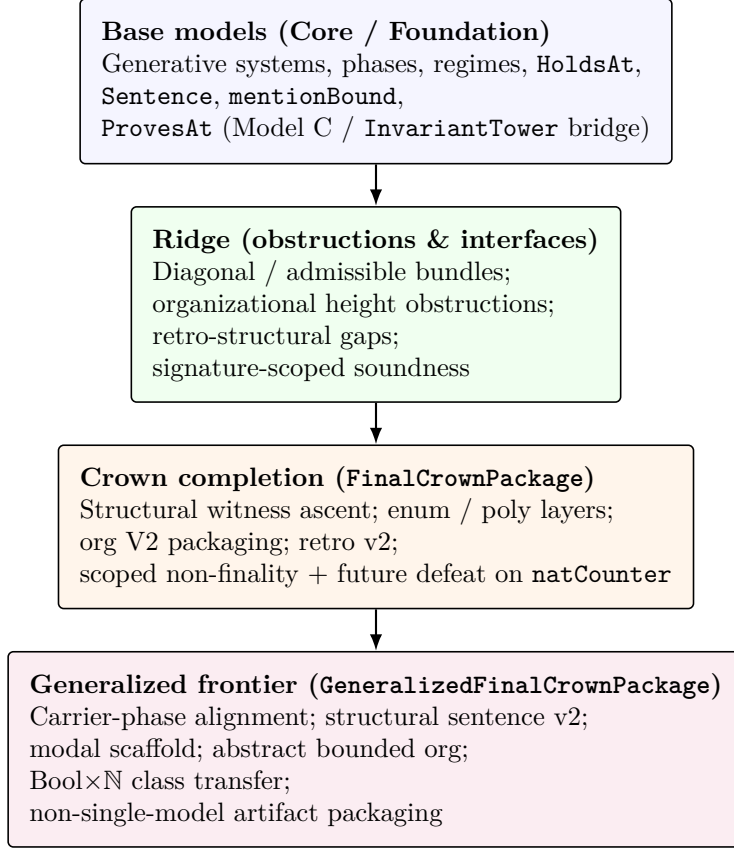


Figure 1: Informal dependency layers for the flagship generalized final package. Arrows mean “mathematically depends on (directly or via re-exports).” This diagram is for exposition only; precise declaration-level edges live in the machine-checked modules.

Table 1 – continued

| Informal claim                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Lean theorem | Assumptions | Boundary |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------------|----------|
| <b>Global obstructions (boundaries, not headline package).</b> There is no uniform diagonal over all raw admissible-interface rows, and no infinite paradigm-shift chain for every singleton-within regime family—these are lemma-level refutations of <i>literal</i> universal glosses; see the “Trust boundary” section in the main text, with representative lemmas such as <code>NoveltyTheory.Ridge.SignatureAdmissibleBundle.trivialInterface_rows_claim_everything</code> in the artifact. |              |             |          |

## B Human-readable dependency layers

## C Appendix-style theorem inventory (dependency-focused)

**Summit stacks (supporting layers).** *[Draft: SummitPackages / OrganizationFrontier / FinalCrownPackage as supporting narrative, not coequal headlines.]*

**Crown completion layer.** *[Draft: enum / poly / org V2 / retro v2 lemmas with representative anchors.]*

**Obstruction lemmas preventing universal glosses.** *[Draft: name representative counterexample interfaces / empty regimes without internal repo paths in external PDF—use only.]*

## References

- [1] Matthew Cook. Universality in elementary cellular automata. *Complex Systems*, 15(1):1–40, 2004.
- [2] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *Proc. 28th Int. Conf. Automated Deduction (CADE-28)*, volume 12699 of *LNCS*, pages 625–635. Springer, 2021.
- [3] Mathlib Community. Mathlib4: The Lean 4 mathematical library. <https://github.com/leanprover-community/mathlib4>, 2024. Pin tracked in repository `lean-toolchain / lakefile.lean`; paper draft aligned with v4.29.0-rc6.
- [4] Nova Spivack. novelty-theory-lean: Fixed generation, stratified provability, and crown packaging in Lean 4. <https://github.com/novaspivack/novelty-theory-lean>, 2026. Machine-checked library; verify URL if repository is renamed or made private.
- [5] Stephen Wolfram. *A New Kind of Science*. Wolfram Media, 2002.